INDIVIDUAL FINAL PRESENTATION · LOGIC TRACK

My Contribution — Debate Judge

We built and tested the logic-track judge — every metric, every version — and annotated debates as ground truth.

1. **Build**
   logic prototype & API
2. **Test**
   judge v1 & v2 metrics
3. **Annotate**
   ground-truth debates

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Before Media day: what I did

From first research to our logic API

1 Foundations & research

- Passed the Claude Code course; ran local LLMs with Ollama
- Literature review: claims & premises, premise support, counterarguments, undercuts, relevance

2 Logic prototype

- Built two modules on the DeBERTa-v3 NLI model, with a Streamlit demo
- **Premise → claim support:** 8 / 10 correct
- **Rebuttal vs undercut:** 6 / 10 correct

3 Logic API

- FastAPI: /score, /winner, /debate + a Streamlit UI
- An LLM (Ollama) extracts claim + premise; they are compared with Change-My-View arguments from Elasticsearch
- Score = coverage × quality; winner = vote of 4 strategies

Next: I worked on the judge's metric design — NLI, ArgQuality, Delta, Votes — explaining each with a toy example, a test bench and flowcharts. This became judge v1 for Media day.

What these first steps showed

Three early findings I carried into the judge

Prototype	“Penguins are birds” → “Penguins can fly”: the model said neutral (0.998) instead of contradiction — NLI only knows what is written, not world knowledge	→ The same NLI weakness came back later in judge v2
Prototype	Rebuttal vs undercut: all 4 mistakes sat right at the boundary, with low confidence (0.53–0.68)	→ The model's confidence shows where it is unsure
API	Scoring by similarity to reference arguments rewards known arguments, not good reasoning — and without Elasticsearch, the LLM wrote the references itself	→ Next step: separate metrics for separate questions

Before: Judge v1 at Media day

Many signals, hand-picked weights, one number

Per argument

NLI

do the premises support the claim?

ArgQuality

similar arguments in the IBM corpus (LLM fallback)

Delta & Votes

did a similar Change-My-View argument change minds / get upvotes?

→ ***weighted composite***

Per debate

Engagement

does the turn answer the opponent?

Information density

does it add new content?

Rebuttal effectiveness

does it contradict the opponent's claims?

+ argument quality over the debate

Emotion

Composure

from the Emotion API

Final score = 0.5 × individual + 0.4 × debate + 0.1 × composure → 0–100 → winner

What I did on v1: tested every metric by hand

7	17	3	✓
endpoints tested	designed test cases	full debates	final formula re-computed by hand

What I found	My suggestion
NLI scored an unrelated claim (“Cats make better pets than dogs”) 0.981	→ **Check topic relevance before NLI**
NLI gave a weak argument 0.999 — it checks the connection, not the substance	→ **Keep a quality metric next to NLI**
An obvious rebuttal was missed on raw paragraphs (just under the 0.55 threshold)	→ **Extract claims first, then compare**
Delta & Votes: the same topic matched one angle but missed another	→ **Say “not applicable” — never a fake score**

Also traced a wrong composure value (always 78 / 70) to the test page's form — not the judge.

Before → After: from many metrics to 5

BEFORE · v1

- About 9 signals
- Hand-picked weights
- One 0–100 score
- No human ground truth
- IBM ArgQ didn't fit our long debate turns

RESEARCH WE USED

- **Wachsmuth et al. 2017** — cogency = acceptability, relevance, sufficiency
- **Pollock 1987, Prakken 2010** — rebut / undercut / undermine
- **DeBERTa-v3 NLI** (MNLI · FEVER · ANLI)

AFTER · v2 final

- **Turn level:** stance support, logical cogency, consistency
- **Debate level:** rebuttal effectiveness, information density
- Each reported separately — no weighted total

Every version re-tested: Aug 24 first v2 → Aug 28 my tests → Aug 31 LLM reasoning added → Sep 8 metrics simplified, consistency its own metric → Sep 27 final → Sep 28 my final tests

Left out of the final 5: ArgQuality, engagement, composure and the weighted total. Delta & votes and argument survival stay only as optional extras.

What I did on v2: tested every version

Early v2 · Aug 28

4 NLI metrics · 2 topics (WFH, sugar tax)

- Cut-off generated text flipped stance: WFH turn 2 went LOW (0.14) → HIGH (0.83) once the broken ending was removed
- CON turns were read as neutral → dual stance (PRO + CON sentence) worked better
- NLI cogency collapsed on a broad claim (relevance 0.008) → cogency later moved to an LLM judgement
- Consistency was the strongest metric (0.98–1.00)

Final v2 · Sep 28

3 turn-level metrics · 5 topics × 6 turns · test script

- **Stance support:** 17/30 wrong — it rewards repeating the side, not arguing for it
- **Logical cogency:** gemma 19/30 vs qwen 12/30 correct — qwen gave the same scores to every turn
- **Consistency:** 18/20 — but it missed the one real change of position

What the testing led to: cogency moved from NLI to an LLM judgement · stance support still needs a new method · consistency needs test cases with known answers

Annotation: building ground truth

To check a judge, you need debates scored by humans

1. **Proposed 3 topics**
   working from home · sugar tax · money can buy happiness

2. **Guideline work**
   split a turn into arguments → claim + premises → link to the opponent; scores 0–3

3. **Annotated 3 debates**
   6 turns each: stance, acceptability / relevance / sufficiency, consistency, rebuttal, information density, winner + reason

4. **Shared topic**
   “money can buy happiness” by all 5 annotators → vote only 3–2, low agreement → guideline revised

Where annotation and testing meet

WFH, turn a-r3 (PRO): *“The key is to leverage the best of both worlds, combining the flexibility of working from home with the social and collaborative aspects of office settings.”*

My annotation: stance 1, consistency 1 — A gives up the motion, so B wins.

The judge: stance LOW only by chance · sufficiency 2 · consistency 0.997 — no metric caught it.

Why it matters: scores can look fine on their own — only human judgement shows what they miss.

What I learned: tools & skills

Ollama

Run LLMs locally and on the Webis cluster (llama3.2, qwen2.5, gemma4). Bigger models answer better but start slower.

Hugging Face

transformers & sentence-transformers: DeBERTa-v3 NLI, MPNet / MiniLM embeddings for similarity.

FastAPI + Pydantic

REST endpoints with clear request / response contracts and automatic docs.

Streamlit

Quick demo UIs for our prototypes.

NLI

Checks if one sentence supports, contradicts or ignores another. Used for premise support, stance & consistency — and saw where it fails.

Claude Code

Agentic coding: plan first, review every change, keep everything in git.

Other concepts: LLM-as-a-judge · argument quality (Wachsmuth) · attack types (Pollock) · inter-annotator agreement

Biggest lesson: a metric can pass easy test cases and still fail real debates — test both, and compare with human annotations.

Difficulties I faced

1. **The metrics kept changing**

   a new version almost every week, so every version had to be re-tested

2. **NLI is hard to predict**

   the same meaning can get a different score, and words like “not” can flip the result

3. **Messy inputs**

   generated debates had cut-off text, and some claims don't make sense on their own

4. **Hidden changes from AI tools**

   fixing bugs with Claude or ChatGPT also changed other code quietly, so it took time to understand